Cover Letter – Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

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To: Prof. Dr. Igor Kuzle, Editor-in-Chief, *Electrical Engineering* (*Archiv für Elektrotechnik*)

Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia

Subject: Submission of Original Research Article (via Springer Nature Transfer Desk):

"PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting"

Dear Prof. Dr. Igor Kuzle and Esteemed Editorial Board,

I am pleased to submit my original research manuscript entitled **"PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting"** for consideration as an **Original Research Paper** in *Electrical Engineering (Archiv für Elektrotechnik)*.

Scientific Context and Alignment with *Electrical Engineering*

Accurate multi-step forecasting of winding hot-spot and top-oil temperatures supports dynamic thermal rating, condition-based asset management, and cooling control for oil-immersed transformers. IEEE Std C57.91 and IEC 60076-7 associate each sustained 6 °C rise above the 110 °C reference with an approximately doubled aging rate.

The work builds directly on the second-generation wavelet forecasting methodology published in this journal by **Bezerra et al. (2024)** (*Electrical Engineering*, Vol. 106, Issue 3, pp. 3017–3029, DOI: 10.1007/s00202-023-02134-5). It advances that line in three respects:

1. **Continuous dyadic Morlet wavelet KAN.** Learnable wavelet responses replace discrete B-spline knot adaptation, avoiding grid-reallocation discontinuities.
2. **Multi-kernel trend-seasonal decomposition and a routing gate.** A neural gate interpolates between a linear projection and the wavelet pathway. Proposition 1 establishes asymptotic linear structural recovery as representational containment: the zero-gate limit contains the multi-kernel linear subspace, not an optimization or test-error guarantee.
3. **Bounded CLPSO warm-start.** A constriction-factor CLPSO searches a four-dimensional space before AdamW fine-tuning and SWA, using 5 particles for 5 generations (exactly 25 evaluations; generation 1 is the initial population).

Summary of Contributions and Evidence

- **Theoretical scope.** Proposition 1 provides a representational-containment result, stated and proved as such rather than as a generalization guarantee.
- **Benchmark evidence.** Across 16 canonical ETT configurations (lookbacks {96, 336}; horizons {96, 192, 336, 720}), PSO-TCWavKAN has the lowest MSE in 7 of 8 ETTh2 configurations. At ETTh2 L=96 it reduces MSE versus DLinear by 6.3%, 6.2%, 8.5%, and 13.6% at H=96/192/336/720. Its overall mean MSE is 1.4% above DLinear because of long-horizon H=720 dispersion, and it wins 10/16 conditions against DLinear and 12/16 against NLinear.

- **Statistical reporting.** The nominal two-sided Wilcoxon test versus NLinear gives W=28.0, p=0.0386; the Holm-corrected value is p=0.116. The rounded variant gives W=29.0, p=0.0443, Holm p=0.133. Neither is significant after correction at alpha=0.05.
- **Ablation evidence.** A1–A4 comprise 192 three-seed runs; A5 is the five-seed benchmark reference, not a separate three-seed run. The ablation is a descriptive comparison across different seed budgets.
- **Noise diagnostic.** The noise study is an exploratory mixed-source, single-checkpoint diagnostic; the manuscript reports raw MSE only and does not use it for a multi-seed robustness claim.
- **Computational profile.** The model has 44,841 parameters at L=96/H=96 and 796.9 K at L=336/H=720. The reported 3.4 ms batch latency and the 3.3x training-speed advantage over the standard B-spline KAN diagnostic (1.24 vs 4.15 s/epoch; PatchTST Lite is 2.85 s/epoch) are observational measurements from the original Kaggle T4 run; the profiling logs are not archived.

Reproducibility and Data Availability

- **Open benchmark data:** the standardized ETT datasets (https://github.com/zhouhaoyi/ETDataset).
- **Public code and result archive:** https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting. The release includes the v9.1 implementation, the aggregate benchmark and ablation CSV matrices, a script that reproduces both the rounded and full-precision Wilcoxon/Holm results, and a parameter-count reproduction script for the four released models in Table 7. The Fig. 3 source-data JSON and plotting script are included; the diagnostic script additionally requires the public efficient-kan package and the ETT CSV files.
- **Release boundaries:** per-seed raw logs, model checkpoints, per-seed noise trajectories, predecessor-diagnostic prediction arrays, and Table 7 profiling logs are available from the author on reasonable request.

Publication Model Declaration

In accordance with Springer Nature policies for hybrid journals, this manuscript is submitted under the **traditional subscription route (without open access fees / 0 APC)**.

Ethical Declarations

- **Originality:** This manuscript represents original research that has not been published previously and is not under consideration elsewhere.
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- **Transfer Notice:** This manuscript is transferred via the official Springer Nature Transfer Desk following editorial recommendation.

Thank you for your time and consideration.

Sincerely yours,

Nguyen Phu Toan

Department of Information Technology

Hanoi University of Business and Technology

29A, Lane 124 Vinh Tuy Street, Vinh Tuy Ward, Hai Ba Trung District, Hanoi, Vietnam

Email: nphu211206@gmail.com | ORCID: 0009-0005-0529-2131 (https://orcid.org/0009-0005-0529-2131)